East West University

Department of Computer Science and Engineering

Summer 2022

Lab Assignment Submission Report-04

ASSIGNMENT TITLE: Writing SQL Statements for Multi-table, Set and Aggregate queries.

Course Code: CSE302

Course Title: Database Systems

Section: 02

SUBMITTED BY

Student Name: Jannatul Ferdous Salma

Student ID: 2020-2-60-005

SUBMITTED TO

Mohammad Rezwanul Huq, PhD

Associate Professor,

Department of Computer Science and Engineering

DATE OF SUBMISSION

18 JULY 2022

LAB TASKS

1)

```
Commit complete.

SQL> SELECT branch_name, branch_city
  2  FROM Branch
  3  WHERE assets>1000000;

BRANCH_NAME     BRANCH_CITY
--------------- ---------------
Redwood         Palo Alto
Perryridge      Horseneck
Round Hill      Horseneck
North Town      Rye
Brighton        Brooklyn
```

Figure:1

2)

```
SQL> SELECT account_number,balance
  2  FROM Account
  3  WHERE branch_name = 'Downtown'
  4  OR (balance BETWEEN 600 AND 750);

ACCOUNT_NUMBER     BALANCE
-------------- ----------
A-101                 500
A-215                 700
A-222                 700
A-217                 750
A-444                 625
```

Figure:2

3)

```
SQL> SELECT account_number
  2  FROM account a, branch b
  3  WHERE b.branch_name = a.branch_name
  4  AND branch_city = 'Rye';

ACCOUNT_NUMBER
---------------
A-333
A-444
```

Figure:3

4)

```
SQL> SELECT l.loan_number
  2  FROM loan l, customer c, borrower b
  3  WHERE b.loan_number = l.loan_number
  4  AND b.customer_name = c.customer_name
  5  AND amount >= 1000
  6  AND customer_city = 'Harrison';

LOAN_NUMBER
---------------
L-17
L-15
```

Figure:4

5)

```
SQL> SELECT *
  2  FROM account
  3  ORDER BY balance desc;

ACCOUNT_NUMBER  BRANCH_NAME         BALANCE
--------------- --------------- ----------
A-201           Perryridge             900
A-333           Central                850
A-217           Brighton               750
A-215           Mianus                 700
A-222           Redwood                700
A-444           North Town             625
A-101           Downtown               500
A-102           Perryridge             400
A-305           Round Hill             350

9 rows selected.
```

Figure:5

6)

```
SQL> SELECT *
  2  FROM customer
  3  ORDER BY customer_city;

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
Brooks          Senator      Brooklyn
Hayes           Main         Harrison
Jones           Main         Harrison
Johnson         Alma         Palo Alto
Adams           Spring       Pittsfield
Lindsay         Park         Pittsfield
Williams        Nassau       Princeton
Curry           North        Rye
McBride         Safety       Rye
Smith           Main         Rye
Majeris         First        Rye

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
Jackson         University   Salt Lake
Green           Walnut       Stamford
Turner          Putnam       Stamford
Glenn           Sand Hill    Woodside

15 rows selected.
```

Figure:6

7)

```
SQL> SELECT c.customer_name
  2  FROM account a, depositor d, customer c
  3  WHERE c.customer_name = d.customer_name
  4  AND a.account_number = d.account_number
  5  INTERSECT
  6  SELECT c.customer_name
  7  FROM loan l, borrower b, customer c
  8  WHERE c.customer_name = b.customer_name
  9  AND b.loan_number = l.loan_number;

CUSTOMER_NAME
---------------
Hayes
Jones
Smith
```

Figure:7

8)

```
SQL> SELECT c.*
  2  FROM customer c, account a, depositor d
  3  WHERE c.customer_name = d.customer_name
  4  and d.account_number = a.account_number
  5  UNION
  6  SELECT c.*
  7  FROM customer c, borrower b, loan l
  8  WHERE c.customer_name = b.customer_name
  9  and b.loan_number = l.loan_number;

CUSTOMER_NAME    CUSTOMER_STR CUSTOMER_CITY
---------------- ------------ ---------------
Adams            Spring       Pittsfield
Curry            North        Rye
Hayes            Main         Harrison
Jackson          University   Salt Lake
Johnson          Alma         Palo Alto
Jones            Main         Harrison
Lindsay          Park         Pittsfield
Majeris          First        Rye
McBride          Safety       Rye
Smith            Main         Rye
Turner           Putnam       Stamford

CUSTOMER_NAME    CUSTOMER_STR CUSTOMER_CITY
---------------- ------------ ---------------
Williams         Nassau       Princeton

12 rows selected.
```

Figure:8

9)

```
SQL> SELECT c.customer_name, c.customer_city
  2  FROM loan l, borrower b, customer c
  3  WHERE c.customer_name = b.customer_name
  4  AND l.loan_number = b.loan_number
  5  MINUS
  6  SELECT c.customer_name, c.customer_city
  7  FROM account a, depositor d, customer c
  8  WHERE a.account_number = d.account_number
  9  AND c.customer_name = d.customer_name;

CUSTOMER_NAME   CUSTOMER_CITY
--------------- ---------------
Adams           Pittsfield
Curry           Rye
Jackson         Salt Lake
McBride         Rye
Williams        Princeton
```

Figure:9

10)

```
SQL> SELECT sum(assets) as total_assets
  2  FROM branch;

TOTAL_ASSETS
------------
    24600480
```

Figure:10

11)

```
SQL> SELECT branch_name, avg(balance)
  2  FROM  account
  3  GROUP BY branch_name;

BRANCH_NAME      AVG(BALANCE)
--------------- ------------
Round Hill                350
Mianus                    700
Perryridge                650
Redwood                   700
Brighton                  750
Central                   850
Downtown                  500
North Town                625

8 rows selected.
```

Figure:11

12)

```
SQL> SELECT branch_city, avg(balance)
  2  FROM account a, branch b
  3  WHERE a.branch_name = b.branch_name
  4  GROUP BY branch_city;

BRANCH_CITY      AVG(BALANCE)
--------------- ------------
Horseneck               587.5
Brooklyn                  625
Palo Alto                 700
Rye                     737.5
```

Figure:12

13)

```
SQL> SELECT branch_name, min(amount) as lowest_amount
  2  FROM loan
  3  GROUP BY branch_name;

BRANCH_NAME      LOWEST_AMOUNT
--------------- -------------
Round Hill                900
Mianus                    500
Perryridge               1300
Redwood                  2000
Central                   570
Downtown                 1000
North Town               7500

7 rows selected.
```

Figure:13

14)

```
SQL> SELECT branch_name,count(loan_number)
  2  FROM loan
  3  GROUP BY branch_name;

BRANCH_NAME      COUNT(LOAN_NUMBER)
--------------- ------------------
Round Hill                       1
Mianus                           1
Perryridge                       2
Redwood                          1
Central                          1
Downtown                         2
North Town                       1

7 rows selected.
```

Figure:14

15)

```
SQL> SELECT c.customer_name, a.account_number
  2  FROM customer c, depositor d, account a
  3  WHERE  d.account_number = a.account_number
  4  AND c.customer_name = d.customer_name
  5  AND balance = (SELECT max(balance) FROM account);

CUSTOMER_NAME   ACCOUNT_NUMBER
--------------- ---------------
Johnson         A-201
```

Figure:15

Learning Outcomes: The following is what we have learned in this lab. We have learned to LIKE operator for String Matching and DISTINCT keyword. We learned how to use the set operation as (UNION, INTERSECT and MINUS). Also learned join operation, ORDER BY clause and aggregate function (GROUP BY clause namely avg, sum, min, max and count). Then finally, we have completed the lab tasks.